

EXPERIMENT – 27

Implementation of RSA Encryption and Dictionary Attack on Small Plaintext Space

CODE:

```
#include <stdio.h>

long long power(long long a, long long b, long long n)
{
    long long r = 1;
    while (b--)
        r = (r * a) % n;
    return r;
}

int main()
{
    long long e = 7, n = 187;
    int message[] = {0, 1, 2}; // A, B, C
    long long cipher[3];

    // RSA Encryption
    printf("Ciphertext: ");
    for (int i = 0; i < 3; i++)
    {
        cipher[i] = power(message[i], e, n);
        printf("%lld ", cipher[i]);
    }
```

```

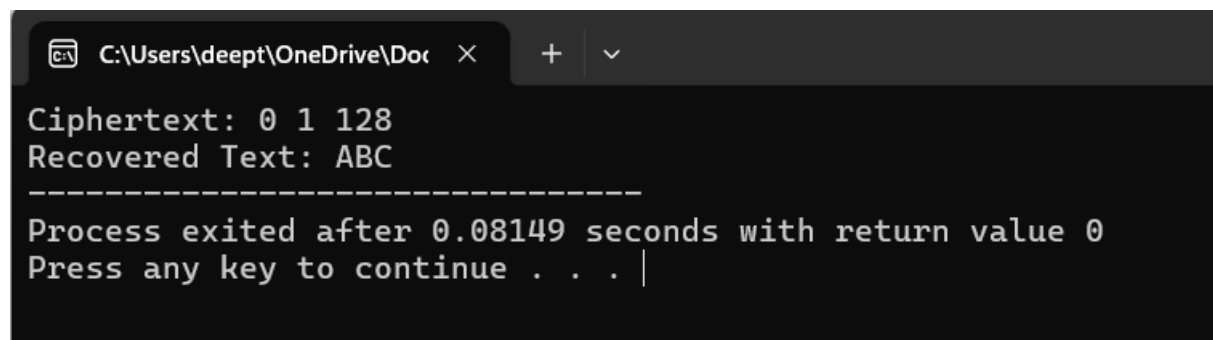
// Attack: Try all 26 possible letters

printf("\nRecovered Text: ");
for (int i = 0; i < 3; i++)
{
    for (int j = 0; j < 26; j++)
    {
        if (power(j, e, n) == cipher[i])
        {
            printf("%c", 'A' + j);
            break;
        }
    }
}

return 0;
}

```

OUTPUT:



```

C:\Users\deept\OneDrive\Doc >
Ciphertext: 0 1 128
Recovered Text: ABC
-----
Process exited after 0.08149 seconds with return value 0
Press any key to continue . . . |

```